IMEN891K / IDSC501 / AXGS-501 · DATA ENGINEERING · FALL 2026

# 1.

# Overview & Logistics

Sangwoo Mo · POSTECH

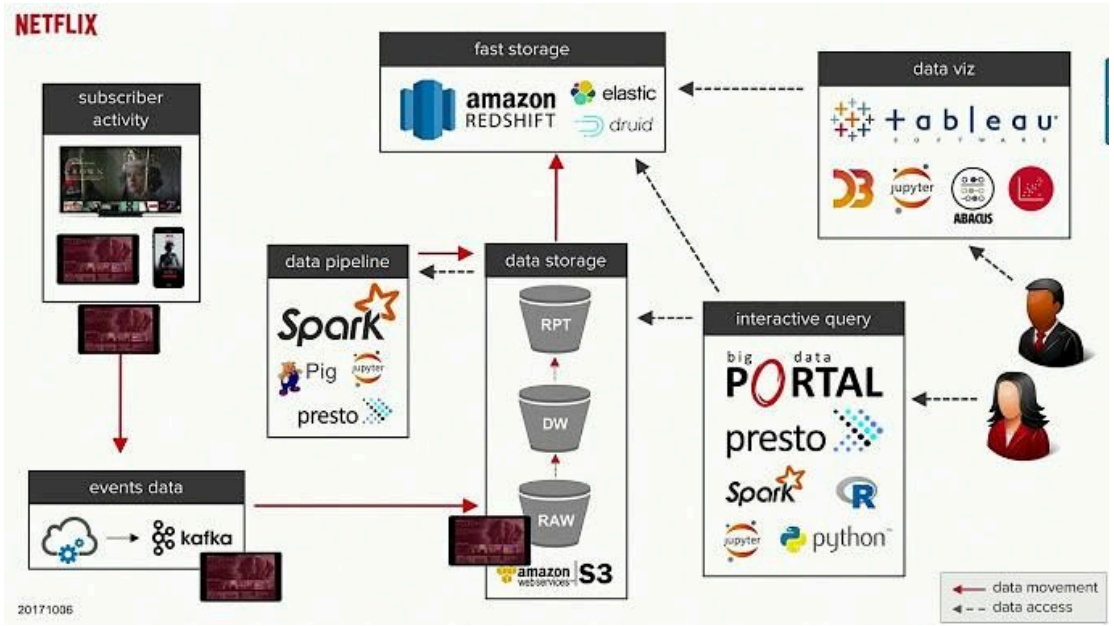
Industrial & Management Engineering

# What is Data Engineering?

Before the definition: how two companies actually run on data

# Case study: Netflix

## One path through the platform




## Follow the data

**Events** enter the pipeline

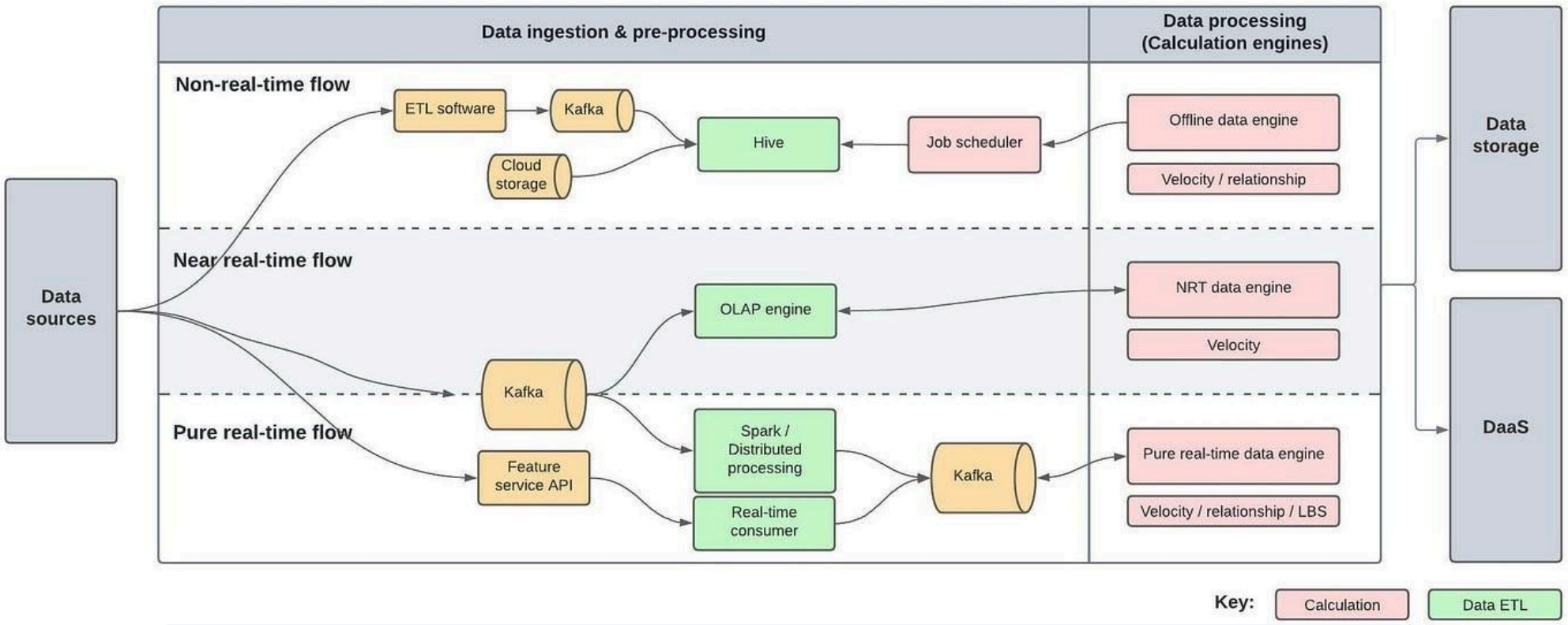
**Storage** keeps records available

**Queries and dashboards** read the prepared data

Follow events through processing and storage to the queries that use them

# Case study: Coupang Eats

coupang eats




Different latency needs get **non-real-time**, **near-real-time**, and **pure real-time** flows

# Outline: Overview and Logistics

| | | |
|---|---|---|
| 01 | **The Pipeline** | follow records from source to outcome |
| 02 | **Course Scope** | collection, storage and monitoring |
| 03 | **Course Logistics** | evaluation, project and AI policy |

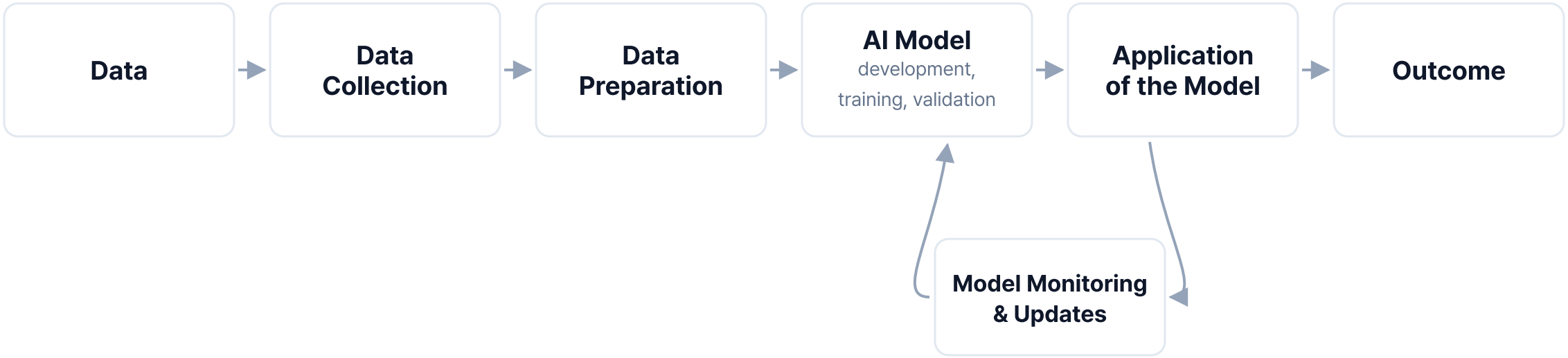
# 01 · The Pipeline


01 **The Pipeline** follow records from source to outcome

02 **Course Scope** collection, storage and monitoring

03 **Course Logistics** evaluation, project and AI policy

# Data Engineering Pipeline

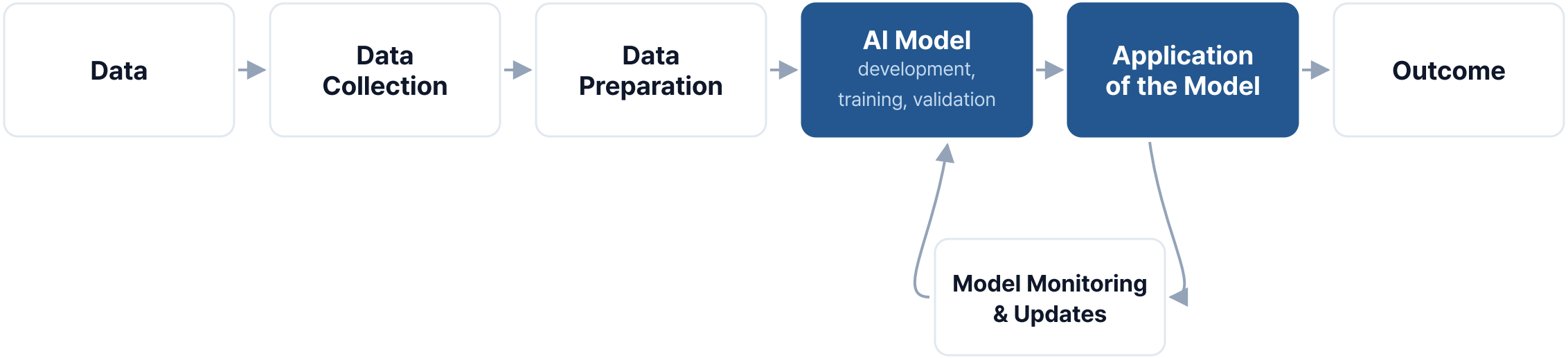



Data becomes an outcome through a **pipeline**, and the model loops back for updates

# 02 · Course Scope

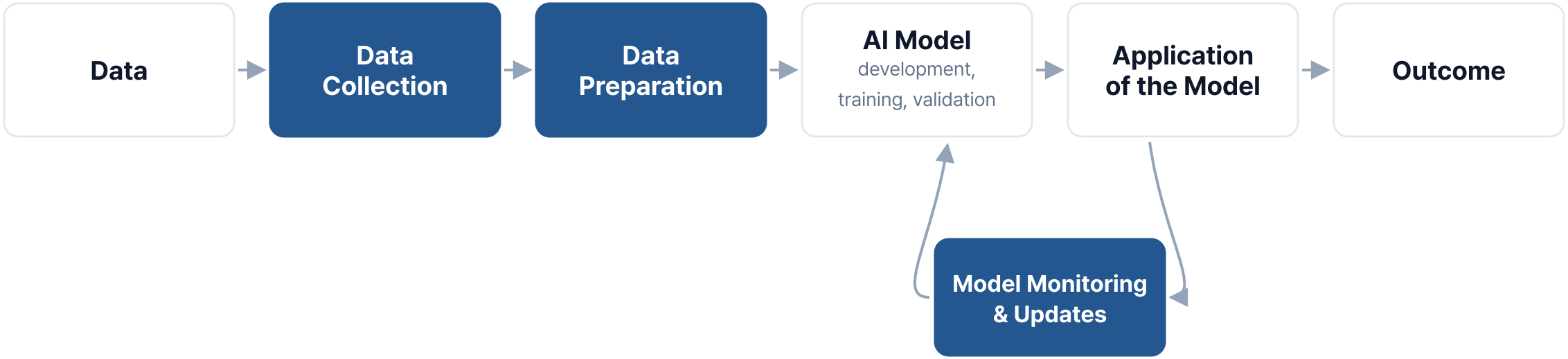
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# Spring Course Scope



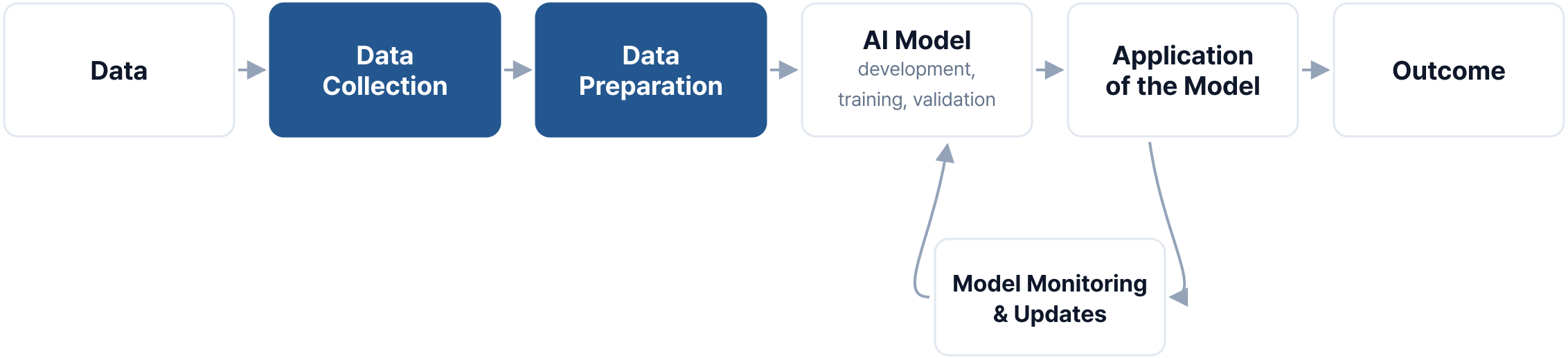

Spring covers **building and applying** the model

# Fall Course Scope




Fall covers what happens **before** and **after** the model

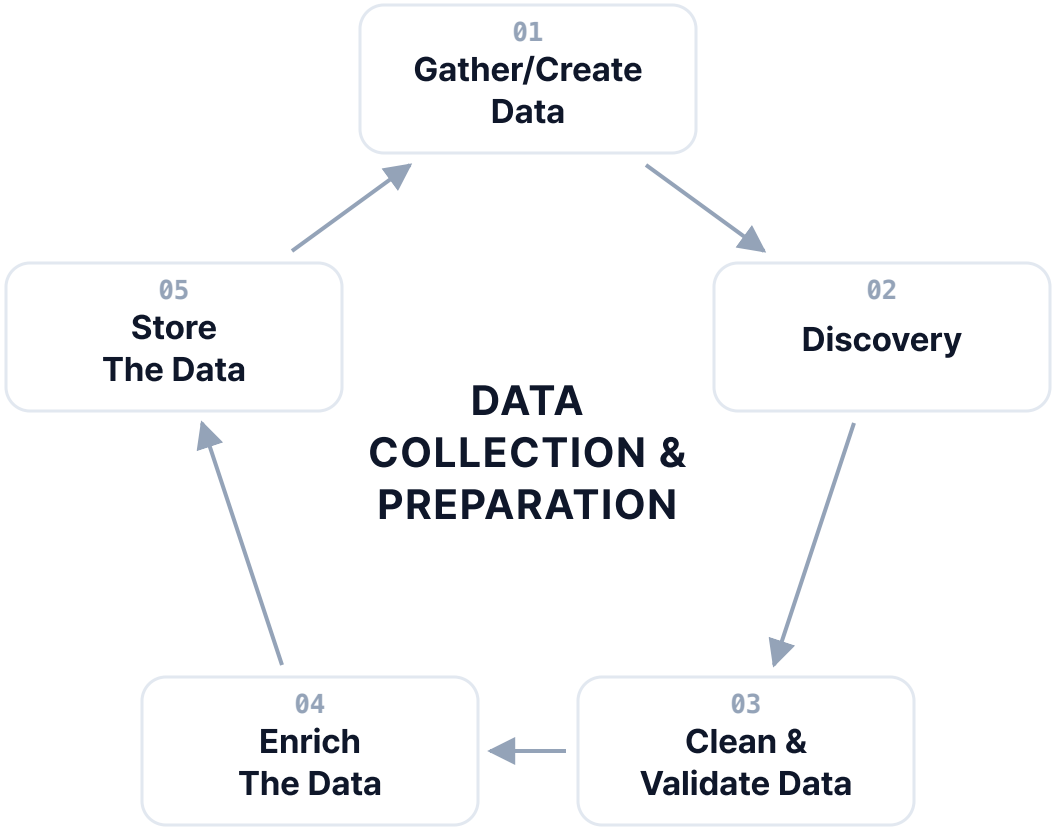
# Part I & II: Before the Model




We start with **data collection** and **data preparation**

# Five Steps Before the Model

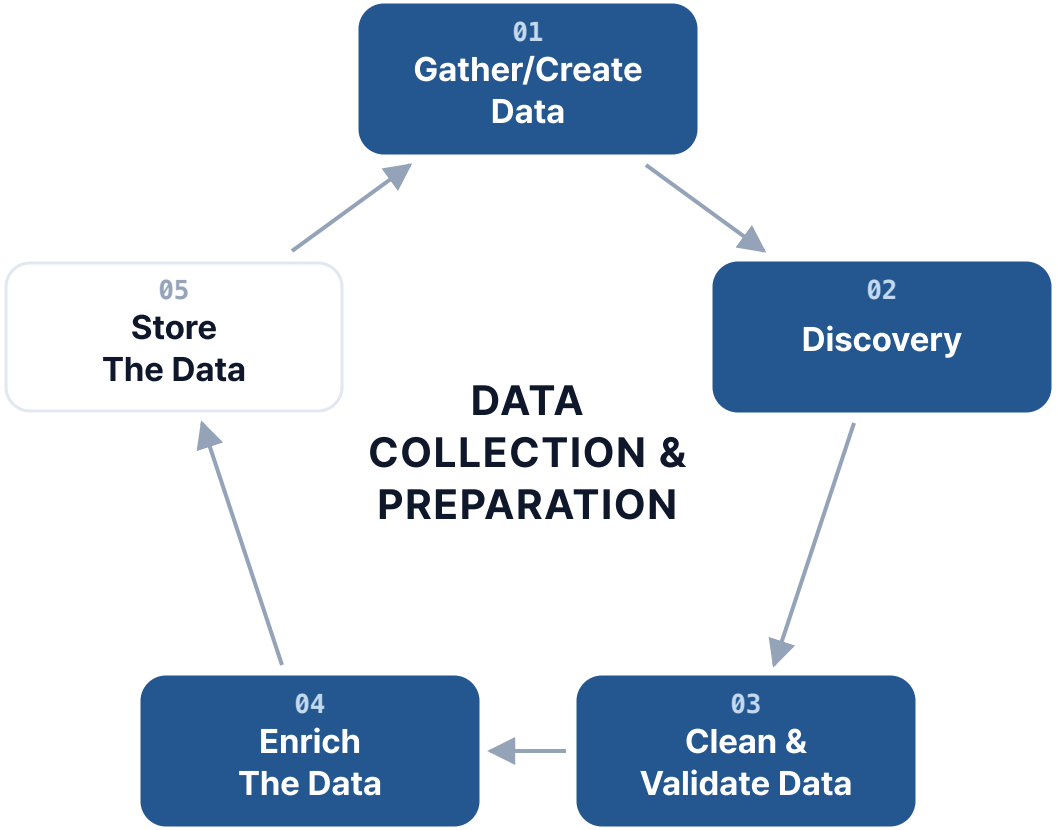
Collection, preparation and storage, as one cycle




Collection and preparation unfold into a **cycle** of five steps

# Part I (Lectures 2–5)

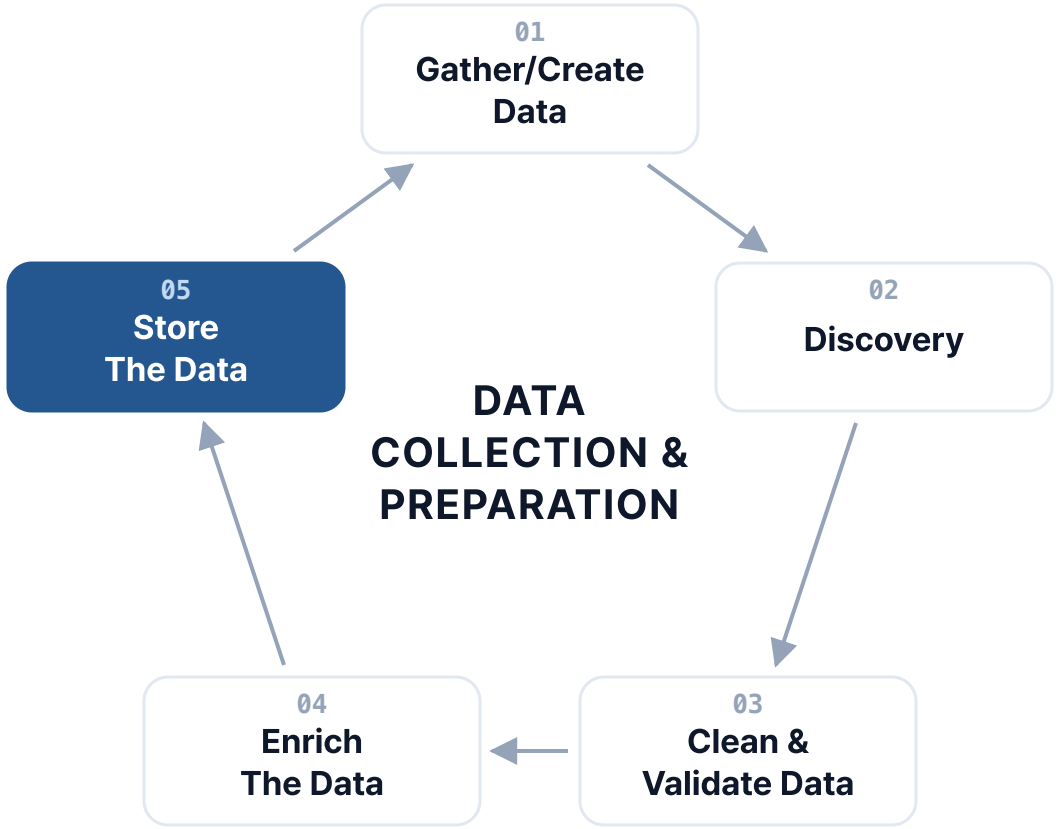
Collection and Preparation




**Part I** covers collection, exploration, and preprocessing

# Part II (Lectures 6–13)

Storage: SQL, NoSQL, Big Data




**Part II** covers step 05: storing, querying and processing the data

# Data Workloads and Tools

The workload determines which storage and processing tools fit

### One local analysis

Read files

Explore and calculate

`pandas · DuckDB`

→

### Shared data service

Repeated queries

Concurrent readers and writers

`SQL · document stores`

→

### Distributed processing

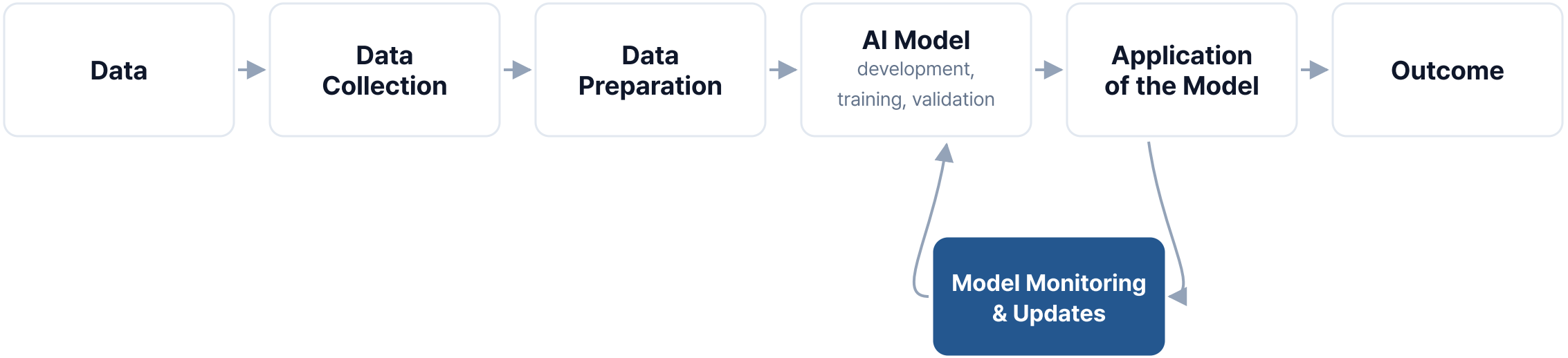
Work across machines

Large batch or event streams

`Spark · stream engines`

Choose tools for the task, shared access and processing requirements

# Part III: Monitoring and Updates




After the model ships, it is **monitored and updated** when needed

# From DevOps to AgentOps

| | DevOps<br>Development + Operations | MLOps<br>Machine Learning Operations | LLMOps<br>Large Language Model Operations | AgentOps<br>Agent Operations |
|---|---|---|---|---|
| What ships | Code | + data, model | + prompts, context | + tools, multi-step plans |
| What fails quietly | Regressions | Data drift | Hallucination, cost | Wrong actions, loops |
| What you watch | Uptime, latency | Accuracy over time | Evals, tokens | Traces, tool calls |

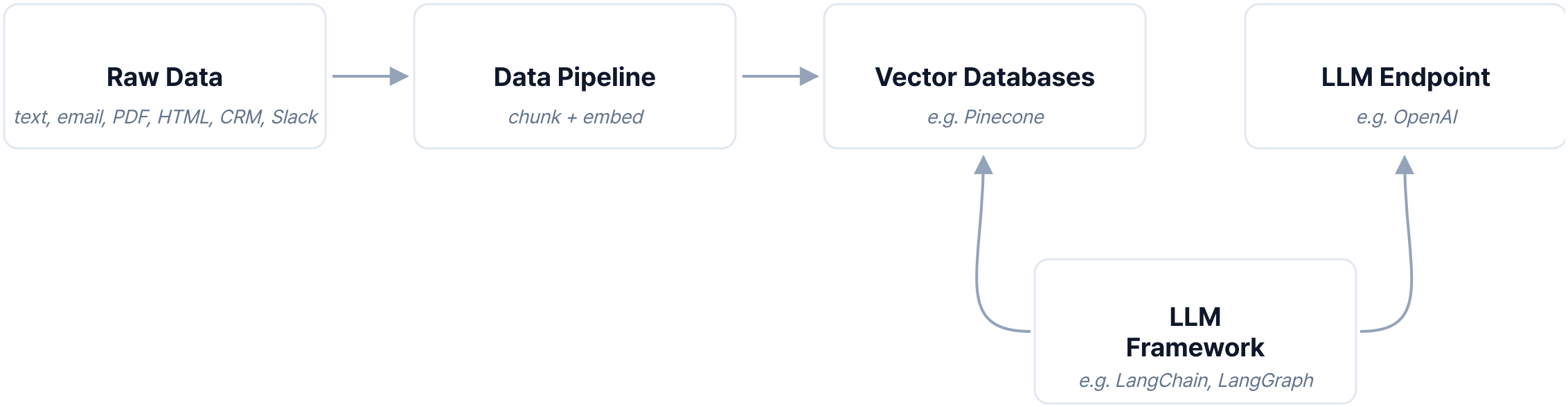
**Part III** walks all four: each step adds what ships and what can fail quietly

# When the Model Is an LLM




The model is now **called**, and the **agent** around it is built

# LLM Applications




In retrieval-augmented generation (RAG), a **pipeline** fills the **vector database** the LLM retrieves from

# Part III (Lectures 14–17)

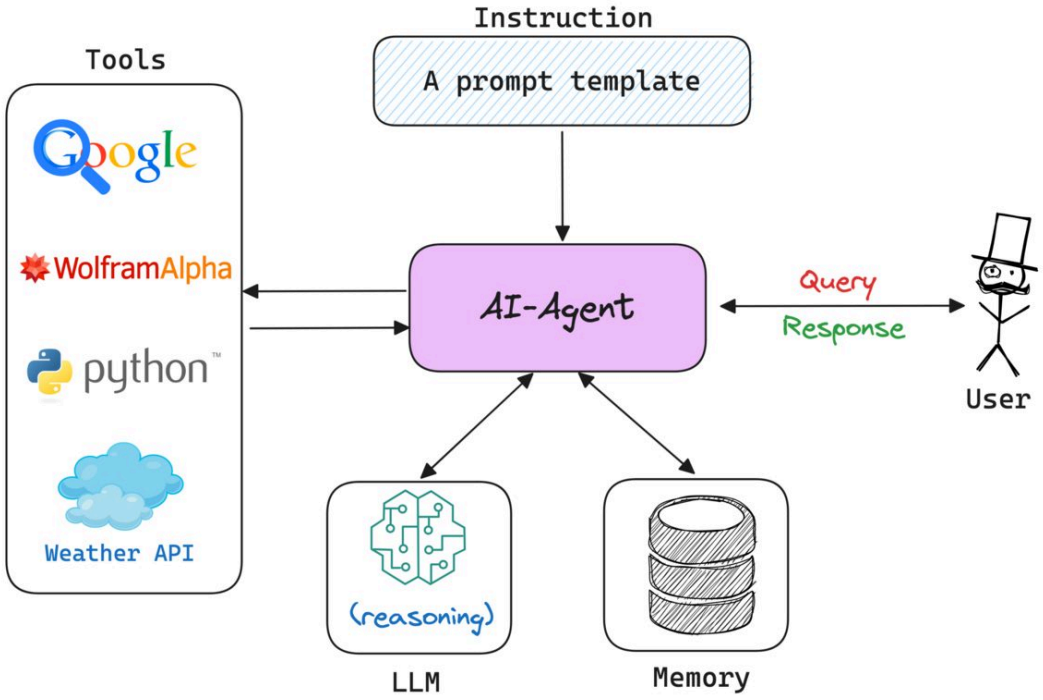
Monitoring and Updates

| | |
|---|---|
| **14 ML Systems** | **DevOps vs MLOps, data drift, model monitoring** |
| **15 LLM Systems** | **LLMs, frameworks, LLMOps, data for LLMs** |
| **16 RAG** | **Retrieval-augmented generation: embeddings, vector DBs** |
| **17 Agents** | **The agent loop, workflows, and agents on your data** |

**Part III** goes from MLOps to agents that work on **your data**

# AI Agent + DB

## Agent structure




## The database role

**Tools** read and act on data

**Memory** retains useful information

**A database** can persist that memory

An agent needs access to data and a way to retain useful information

# Data Engineering Lifecycle

Stages organize the work; deployed systems feed new observations back

### Generate and ingest

Events become records

Collection · Lec 2

→

### Store and transform

Query usable data

Preparation · Lec 3–5

Storage · Lec 6–13

→

### Serve and update

Use data in a system

Monitoring and updates

Lec 14–17

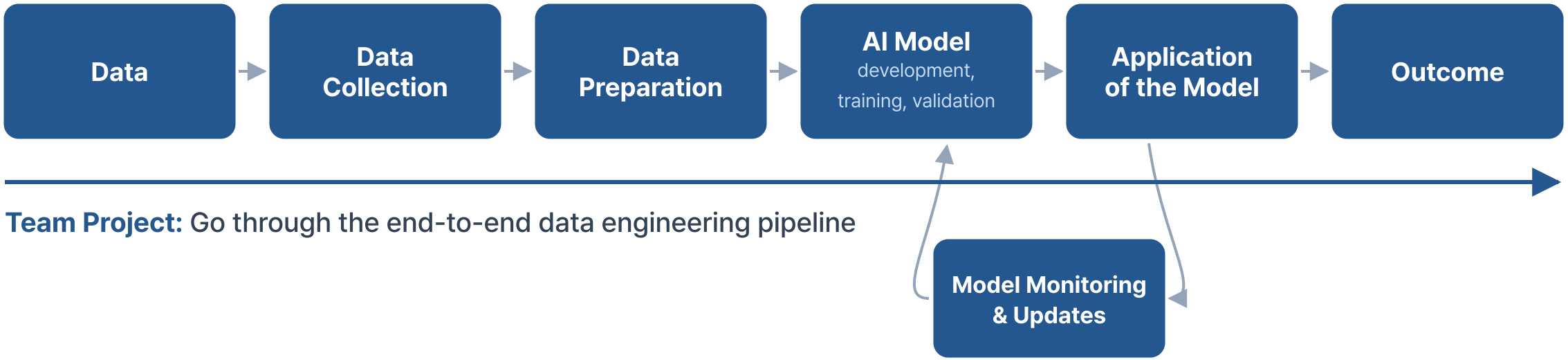
Records pass through stages and return as feedback

# Requirements Across the Lifecycle

| Requirement | Question across stages | Lecture examples |
| --- | --- | --- |
| Security | Who may read or write these records? | Collection · Agents |
| Data management | What do records mean and how do they change? | Storage · Agents |
| DataOps and orchestration | Can a failed job be checked and rerun? | Data Warehousing · Systems |
| Architecture | Where should storage and processing run? | NoSQL · Big Data |
| Software engineering | Can changes be tested and maintained? | A requirement throughout |

These requirements apply across the stages, not just at one endpoint

# By the End of the Class




**Team Project:** Go through the end-to-end data engineering pipeline

By the end, you will have walked the **whole pipeline** once

# 03 · Course Logistics

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# Grading Policy

**40%**
**Midterm Exam**

**20%**
**Project (Proposal)**

**40%**
**Project (Final)**

**0%**
Attendance (excessive absence means F)

Office Hours: **Tuesdays 11:00 am–12:00 pm, Room 319**

# Team Project Evaluation Criteria

| CRITERION | WHAT WE LOOK FOR |
| --- | --- |
| **Novelty** | originality and creativity of the idea |
| **Impact** | relevance and potential usefulness of the solution |
| **Technology** | effective use of the techniques and tools covered in class |
| **Presentation** | clarity and professionalism of the report and presentation |

Team Size: **≤ 5**

# Team Project Milestones

**PROPOSAL · MID-SEMESTER · 20%**

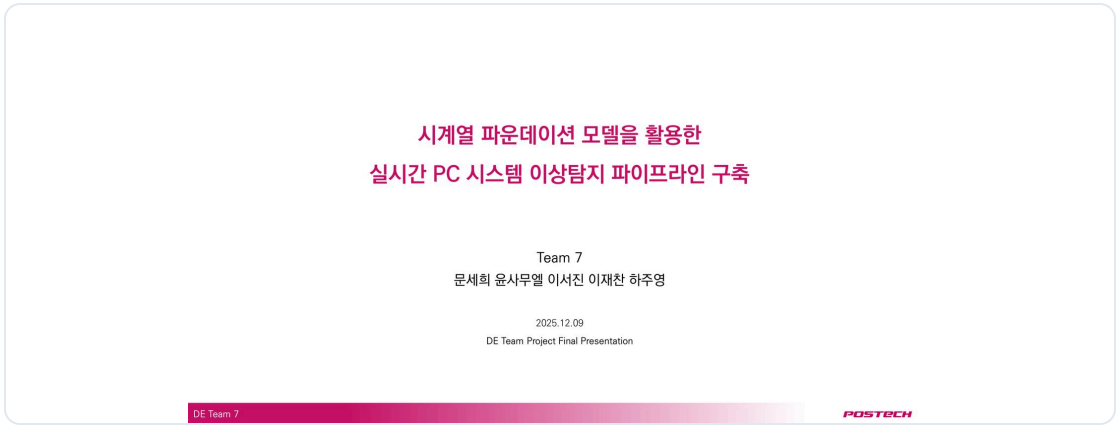
Ideation, data collection & preprocessing, DB management

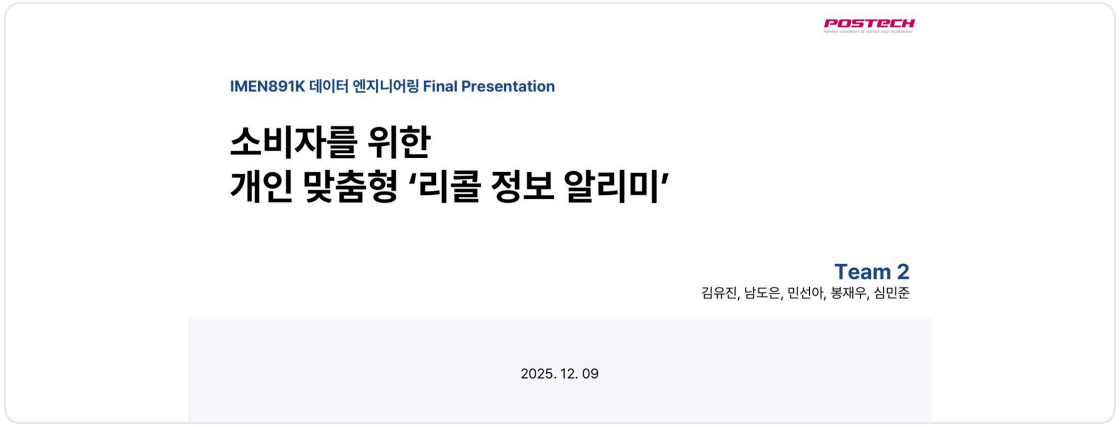
**FINAL · END OF SEMESTER · 40%**

Develop a product that addresses a real-world problem using a large-scale DB (with potential LLM integration)

# Team Project Examples







Four final presentations from **Fall 2025**

# Data Sources for Team Projects

**Public data platforms**

data.go.kr, Kaggle, Hugging Face datasets

**Open APIs**

Government, map, finance, or social platform APIs

**Web crawling**

Pages you collect yourself, within the site's terms

**Your own data**

Lab, company, sensors, surveys you run

**Generated data**

Synthetic or LLM-generated data, with a clear reason

**Or a mix**

Combine sources; joining them is part of the work

Any source is fine if you are **allowed to use it**: check the license, terms, and personal data

# AI Policy



ChatGPT

Do not **compete** with AI, **utilize** it!

# Takeaway: Data Engineering

- Define **what the record means** and whether it supports the task
- Choose storage and processing for **the workload and required guarantees**
- Check results and failures; use **observed outcomes** to guide updates